

Econometrics I

Lecture 1: What is Econometrics?

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The Concept of Ceteris Paribus

The impact of the variable on another while holding everything else fixed. If we change one of the independent variables while holding all other ones fixed, then assess how the dependent variable changes Causal Relationship.

Example: The demand function

$$Q^D = f(P, \text{determinants})$$

This diagram represents the delta in Q^D when only P changes while holding all demand determinants fixed.

Ceteris Paribus (all other factors fixed) Q^D is inversely related to its own price.

$$Q^D = f(P, P^c, P^s, I)$$

where:

P – own price

P^c – price of a complement

P^s – price of a substitute

I – income

$I \uparrow$

In all of these exercises, we are applying the concept of Ceteris Paribus.

What is Econometrics?

- Aim is to measure in a quantitative manner economic relationships among economic variables.
- We need economic data to do this.
- Ex. Suppose we are interested in measuring the effect of one variable, say X , on another variable, say Y .
- Try to estimate the change, in the average value of Y due to a change in one unit of X .
 - Since economic data on X, Y would come from real-world human behavior, and therefore from the point of view of the econometrician/researcher this data is random.
 - Appeal to Probability/Statistics.

This framework leads naturally to the concept of Conditional Expectation: $E(Y|X)$

Example: A small town with 10000 households,

Each household:

- X – annual income (thousands of dollars)
- Y – annual savings

Then

- $E(Y|X = 10)$?
- $E(Y|X = 15)$?
- $E(Y|X = 20)$?

We can perform this exercise for all values of income x :

$E(Y|X = x) = f(x) :=$ Regression function of Y on X

Notation: Random Variables are in capital letters. Realizations of R.V.'s are in lower-case letters.

Example (continued)

$$Y \leq X$$

$$X \sim \text{Gamma}(2, \lambda), \lambda = 0.7 \quad Y|X = x \sim \text{Uniform}[0, x]$$

We will use R, to simulate 10000 observations on X, Y .

Example (continued)

x_0	Size	$E(Y X = x_0)$	Increment ΔY	Effect ($\frac{\Delta Y}{\Delta X}$)
2.5	403	1.25368	-	-
5	135	2.60486	1.35118	0.540472
10	4	4.85862	2.25376	0.450752

Effect $\approx 0.5 \rightarrow$ effect is constant

A change in income X of 1 unit (1000) yields on increase in savings of 0,5 (500).

$\rightarrow$ Propensity to save given your income is 0.5

In general the effect of X on Y would depend on other variables.

In the previous example, the effect of X on Y was "constant", actually we expect it to be different given the value of X .

For example: for each household savings can depend:

- Income
- Family size
- Education of the Head of the Household

If we are interested in estimating the effect of income in savings we need to find a way to control for those other variables.

→ Ceteris Paribus Analysis.

Say we have data on a set of k variables that describe the household: $X_1, X_2, \dots, X_k \rightarrow$
explanatory variables OR independent variables OR regressors OR controls

$Y \rightarrow$ dependent variable OR explained variable OR regressand

We would have to consider the conditional expectation of Y on $X_1, X_2, \dots, X_k$

$$h(X_1, X_2, \dots, X_k)$$

regression function of Y on $X_1, \dots, X_k$

Functional Relationship exists between Y and $X_1, \dots, X_k$. This relationship is causal, for any values $V_1, \dots, V_k, h(X_1, X_2, \dots, X_k)$

Regression function $h(X_1, X_2, \dots, X_k)$

- Comes from Economic Theory.
- Represents the results from an Economic Model.
 - Equilibrium equation from the analysis of a model.

Difference between Causality and Spurious Correlation:

- Causality: each X **causes** a particular value of Y_v
 - $X_1 = V_1$
 - $X_2 = V_2$
 - $X_k = V_k$
- Spurious Correlation
 - positive or negative correlation – not necessarily causal
 - other disciplines only look at correlation,

Econometrics looks at Causality!

Econometrics, nevertheless, can be used

- Forecast (prediction)
- Asses some relationship (not necessarily causal) between variables

Here Econometrics intersects with Machine Learning, which

- looks purely at data (big data) at which some techniques are applied, to forecast and look for patterns
- is not looking necessarily at causality

Regression function $h(X_1, X_2, \dots, X_k)$

Suppose again Y =savings, X_1 =income, $X_2, \dots, X_k$ =other determinants of savings

We are asking what is the ceteris paribus effect of income on savings?

Formally, we are interested in estimating:

$$\frac{\partial h(X_1, X_2, \dots, X_k)}{\partial X_1} \Big|_{X_1 = v_1, X_2 = v_2, \dots, X_k = v_k}$$

The above is a very general formulation of the ceteris paribus effect of X_1 on Y

Steps in an Econometric Study

- ➊ Formulating the question(s) of interest, usually in terms of what effects are to be estimated.
- ➋ Application of the relevant theory to define the pertinent variables $Y, X_1, \dots, X_k$, and gather the corresponding data.
- ➌ Define the functional relationship of interest; or at least define the signs. of the ceteris paribus effects.
- ➍ Estimation of the model.
- ➎ Make some verification checks on the model.
- ➏ If the model passes the above checks, use it to estimate the effects of interest.

Example: What is the effect of education on the wage at a firm?

This question has to do with the economics of personal growth.

From step 2, some relevant variables are:

Y = average hourly wage of a person.

X_1 = years of education

X_2 = years of experience

X_3 = gender

X_4 = years of permanence at the firm

etc.

$$E(Y | X_1 = x_1, \dots, X_k = x_k) = h(x_1, \dots, x_k)$$

$$\text{C.P. eff. } \frac{\partial h}{\partial x_j}(x_1, \dots, x_k)$$

Economic Theory already suggest a possible functional relationship between the variables. In the next lecture we will discuss the conditional approach, based of conditional expectation and the Best Linear Predictor of Y in terms of $X_1, \dots, X_k$.

$$\hookrightarrow h^*(x_1, \dots, x_k) = a_1 x_1 + \dots + a_k x_k$$

$a_1, \dots, a_k$ are constants

The functional relationship that emerges from step 3 will usually depend on unknown constants that we will call parameters. This is called the parametric approach. The statistical technique used is called point of parameter estimation → Method to estimate these parameters, which is step 4, is one of the main topics of the course. A basic check on the model (step 5) is to compare the estimated signs of the ceteris paribus effects with those from the theoretical signs. Another important check is to see if we have outliers in the data because they can affect the magnitude of the estimated effects.

Typically, the model requires too many statistical assumptions to be useful.

- Conditional Approach is too restrictive.
 - A lot of real-world data do not satisfy the assumptions needed for this approach to be fully valid.
- Use asymptotic (what happens when we have a large sample) approach which requires milder assumptions that support the statistical properties of the estimations of the parameters.

Finally, if the checks preformed do not show strong evidence against the model, it can be used for inference of the effects of interest.

Note: If we do not have data on an explanatory variable which is important in the relationship of interest, that may cause bias on the estimator of the parameters in the model.

→ Important aspect in step 2 whose failure could be evidenced in step 5.

Types of Data

1. Cross section data

- Typically comes from a sample from a population at a certain point in time.
Example: A survey of different households.
 - A very important characteristic of cross-section data is that the responses given by each observation/element/unit is independent of one another, observations and their values are independent between units
- statistical independence.

$$[Y_i \ X_{i1}, \dots, X_{ik}]_{i=1, \dots, n}$$

→ row in the matrix of observations **rows are independent** $X_1, \dots, X_k$ are the **columns that are not necessarily independent.**

2. Time Series

- Across time, over a period years/months/days etc.
- In regular intervals

Example:

- stock prices over time (sometimes microseconds)
- CPI (consumer price index)
- GDP
- unemployment

$$(Y_t, Z_t, \dots, X_t)_{t=1, \dots, T}$$

- Rows or observations across time are not independent

$$unemployment_t, GDP_t, inflation_t$$

→ $unemployment_{t-1}$ is not independent of $unemployment_t$

3. Pooled-Cross Section

- All characteristics of a cross-section
- It pools different samples from times/regions
- Each sample at each year may be different (individuals, firms, households, etc.), BUT, it is the same variables.
- Rows are independent.

4. Panel Data

- A combination of a cross-section with a time series.

$$[Y_{it} \ X_{it}, \dots, X_{kt}]_{i=1, \dots, n; \ t=1, \dots, T}$$

→ Observations are  independent across-time and possibly not even across units.

Example: If t denotes a region, there may be same relationship between Y_i and Y_j , $X_{itk}, \dots, X_{itk}$ and $X_{jtk}, \dots, X_{jtk}$ for a fixed t and all i and j